Normalized Augmented Inverses on Finite Contours

Scoped Ram Refinements for Tangent-Flow Uniqueness

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Abstract

The assigned manuscript proposes fixed-frame uniqueness of all backward tangent flows at the first singular time of a smooth closed embedded surface in three-dimensional Euclidean space. Its normalized stationary construction combines a Gaussian Jacobi inverse, compact observations, complex end contours, finite opening jets, a nonzero action jump, and a localized gradient inequality. We retain that global dependency chain and prove narrower improvements at its finite-contour interface. A sharp disk calculation quantifies the accretivity lost when dividing by the pulled-back drift, including its chain-rule term. An indefinite pairing construction enlarges a prescribed observation space by exactly its radical dimension and balances the added Gram block. Full augmented Schur residual identities control inexact additional observations without deleting their sources or multipliers. Finally, the manuscript's mixed-gauge estimates imply a preliminary weighted tolerance of order B^{-p} for $p > \alpha + 1$, where $\alpha > 2.1$, instead of fixing $p = 4$. This last extension applies only to fixed-radius preliminary contours and does not transfer the action-jump conclusions to the enlarged data class. Domain realization, two-sided solvability, natural exit traces, and the order of large-scale limits remain explicit. Contribution assessment: C3/5, source-specific local refinements with complete proofs and a larger preliminary tolerance class, while the global uniqueness theorem and its analytic dependencies remain inherited.

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1 Problem, Scope, and Source Status

Let $M_0 \subset \mathbb{R}^3$ be smooth, compact, connected, embedded, and without boundary. Let M_t, $0 \le t < T$, be its smooth mean curvature flow up to the first singular time. At a fixed singular point (x_0, T) define

$$M_s^\lambda = \lambda(M_{T+\lambda^{-2}s} - x_0), \qquad s < 0. \tag{1}$$

The assigned source [1] asserts that the entire family as $\lambda \to \infty$ converges locally smoothly with multiplicity one to $\sqrt{-s}\,S$, uniformly for s in compact subintervals of $(-\infty, 0)$, in the original ambient frame. No rotation depending on the sequence is allowed. Its additional assertion about every slice of a backward integral Brakke representative uses local mass and curvature integrability, the Brakke inequality with every pair of endpoints, and the stated spacetime weak-limit premise. It is not a claim about post-singular continuation.

The local theorems proved here do not establish (1). They apply to the precise augmented operators and finite contour domains described below. Statements called *source inputs* are inherited from the indicated proofs, even where their mechanism is reconstructed. No finite matrix, sampled contour, or small residual is treated as a global mean curvature flow proof.

1.1 Pinned manuscript and actual reading

The preserved row is M683, family 355, with source directory

`preprints/Tangent-flow-uniqueness-2026-09-24/build`.

The preserved repository revision is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. All fourteen source TeX files and the bibliography were literally read: the main paper and the introduction, geometry, linear, finite-linear, gauges, real-comparison, contours, arrays, jump, phase, finite-jet, localized, and flow sections. The accompanying local ledger records exact paths, line scopes, and SHA-256 hashes. The captured October 8 base/target tree check is byte-identity evidence, not a new proof check.

1.2 Current primary context

The public primary check on October 8, 2026 distinguishes established special cases. Schulze treats a closed smooth multiplicity-one tangent shrinker [6]; Colding–Minicozzi treat cylindrical uniqueness, including axis uniqueness [7]; Chodosh–Schulze treat multiplicity-one asymptotically conical tangent flows [8]. Bamler–Kleiner's current arXiv version is v3 of November 12, 2024 and states multiplicity one for surface flows [5]. Multiplicity one is an input to, not by itself a proof of, the full uniqueness assertion here.

Zhu's current v2 of September 16, 2024 concerns a class of cylindrical shrinkers in arbitrary codimension [9]. Ghosh's cylindrical paper is v3 of July 29, 2026 [10]. Bamler–Lai's v2 of March 21, 2026 gives a PDE–ordinary-differential-inequality approach in the cylindrical setting [11]. These are related alternatives, not automatic replacements for the source's arbitrary finite-ended stationary analysis. Haslhofer's published ICM 2026 account still labels general tangent uniqueness as Conjecture 5.2 [12]. Its survey status and the September manuscript's claim must be distinguished chronologically.

The live repository head returned by the public API was `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The direct family-number documentation URL returned 404, and a root recursive-tree response was truncated. Neither observation proves the absence of a formalization. No Lean kernel, comparator, or end-to-end formal proof was run for this report. Public repository assertions, source-byte identity, the mathematical arguments below, and kernel verification are separate kinds of evidence.

2 Domains and the Actual Normalized Problem

2.1 Gaussian realization and compact normalization

For a source tangent shrinker S put $d\mu_G = e^{-|X|^2/4} d\mu_S$ and

$$L = \Delta_S - \tfrac{1}{2} X^\top \cdot \nabla + \tfrac{1}{2} + |\mathrm{II}|^2.$$

The source realizes L on the Gaussian H^2 completion of compactly supported smooth functions. Its weighted moment argument controls $|X|u$, $|X|\nabla u$, and $|X|^2 u$. The half-density transformation $Uu = e^{-|X|^2/8} u$ gives

$$-ULU^{-1} = -\Delta_S + \frac{|X|^2 + |X^\perp|^2}{16} - 1 - |\mathrm{II}|^2. \tag{2}$$

Properness and the confining potential give the source's compact-resolvent realization. With $K = \ker L$, a finite-dimensional space, select real compactly supported χ_i and set $P_i u = \langle u, \chi_i \rangle_G$. The augmented map is

$$\mathcal{A}_0 = \begin{pmatrix} L & P^* \\ P & 0 \end{pmatrix} : H^2_G(S) \oplus \mathbb{R}^m \longrightarrow L^2_G(S) \oplus \mathbb{R}^m. \tag{3}$$

The kernel-pairing matrix must be invertible; arbitrary independent observations do not suffice. The source also enlarges observations to capture prescribed compact variations. Section 4 makes this enlargement minimal in a precise algebraic sense.

For the nonlinear family the equation is not simply $\Phi(u) + P^*\lambda = 0$ with fixed columns. The compact columns contain the graph area and normal-speed factors needed for first variation:

$$\Phi(u) + \sum_i \lambda_i \chi_i(u) = 0, \qquad P(u) = q, \qquad dF = \lambda \cdot dP.$$

At a nonzero base multiplier, the linearization includes $\sum_i \lambda_i D\chi_i(u)$. Discarding this compact term changes the operator. Finite parameter analyticity is in prescribed real spatial charts; it is not permission to evaluate arbitrary smooth spatial functions at complex points.

2.2 Cylindrical, conical, and boundary norms

On a cylindrical end use $t = \log z$, $E \asymp z^{-2}$ and fixed angular order $h \geq 4$. The model is

$$L_{\mathrm{cyl}} v = v_t - 2e^{-2t}(v_{tt} - v_t) - v_{\theta\theta} - 2v.$$

Its slow rates are $2 - j^2$. A strip domain norm, equivalent to the source's square-sum convention, is

$$N_E(v) = \|v\| + \|v_t\| + \|v_{\theta\theta}\| + \|E v_{tt}\| + \left\| \sqrt{E}\, v_{t\theta} \right\|, \tag{4}$$

with bulk norms in $L^2_t H^h_\theta$. Only $v, v_\theta, \sqrt{E}\, v_t$ are used as first traces. In particular there is no free second-derivative trace. For Fourier coefficients of boundary data the natural norms are

$$\|\varphi\|^2_{T^D_E} = \sum_j (1 + j^2)^h (1 + j^2)(1 + \sqrt{E}|j|)|\varphi_j|^2, \tag{5}$$

$$\|g\|^2_{T^N_E} = E \sum_j (1 + j^2)^h (1 + \sqrt{E}|j|)|g_j|^2. \tag{6}$$

For prescribed smooth data, $\|g\|_{T^N_E} \leq C\sqrt{E}\, \|g\|_{H^{h+1}}$; a trace of a solution needs only its natural norm.

On conical ends the distinct domain includes bounded H^{h+1} height and H^h first derivative, $\mu^{-1} v_t \in L^2 H^h$, and $\mu D^2 v \in L^2 H^h$, where $\mu = e^{-t}$. Its source norm is $\left\| \mu^{-1} f \right\|_{L^2 H^h}$. Substituting the cylindrical norm would lose the two energy identities used in the conical proof. On the core use the source's angularly commuted H^2 domain and L^2 range. These are not an unqualified isotropic high-order Sobolev pair.

2.3 Uniform estimates precede surjectivity

The source's finite-linear proposition assumes a single ordered real outward coordinate, compatible transitions, normalized coefficient bounds, and stopped properly elliptic Fredholm problems of index zero. Its weighted estimate holds on an entire quadrant $B \geq B_0$, $T \geq T_0$ before the conical stop is exhausted. The exterior forward estimate has a strict weight buffer above propagation; otherwise approximate kernels could escape along an end. A compact-limit contradiction then invokes (3). This proves an a priori estimate and injectivity. Index zero on each fully stopped domain supplies surjectivity separately. A homotopy to a coercive real problem supplies that index, not a uniform inverse along the homotopy. Only after the weighted bound is proved may the weights be converted into a physical polynomial loss in B.

The private TR8 realization [3] gives the appropriate caution. For a positive Hermitian divergence-form operator on $H_0^1(\Omega)$, coercivity gives a weak inverse $H^{-1} \to H_0^1$. Its L^2 operator domain consists of functions whose form is represented by an L^2 forcing; H^2 regularity needs additional boundary and coefficient hypotheses. Boundary lifts contribute their own forcing, and a time-dependent lift in a heat problem contributes both $-\partial_t \ell$ and $-L\ell$. Here the Gaussian Jacobi operator is indefinite and the end contours are complex. The positive self-adjoint heat-semigroup proof cannot simply be copied across. The reusable principle is exact domain and source equivalence, not transferred coercivity.

3 A Sharp Drift-Normalization Margin

Consider, on a prescribed real interval, the scalar expression

$$Lv = v_t - a v_{tt} - 2b v_{t\theta} - c v_{\theta\theta} + d_1 v_t + d_2 v_\theta + d_0 v. \tag{7}$$

Let $t = t(s)$ be a prescribed complex path, $H = t_s$ a nonzero function independent of θ, and write all coefficients after pullback. The chain rule gives

$$v_t = H^{-1} v_s, \qquad v_{tt} = H^{-2} v_{ss} - H_s H^{-3} v_s.$$

Multiplication of $Lv = f$ by H yields

$$\delta v_s - \frac{a}{H} v_{ss} - 2b v_{s\theta} - H c v_{\theta\theta} + H d_2 v_\theta + H d_0 v = H f, \qquad \delta = 1 + d_1 + \frac{a H_s}{H^2}. \tag{8}$$

Thus the normalized principal coefficients are $a/(H\delta)$, b/δ, Hc/δ, and the source is Hf/δ. Outgoing data are $g = v_s = H v_t$, not the old coordinate derivative.

Lemma 3.1 (Exact disk margin)**.** *For $z \in \mathbb{C}$ and $0 \leq e < 1$,*

$$\min_{|\delta - 1| \leq e} \operatorname{Re} \frac{z}{\delta} = \frac{\operatorname{Re} z - e|z|}{1 - e^2}. \tag{9}$$

Consequently if $\operatorname{Re} z \geq m\omega$, $|z| \leq C\omega$, $\omega > 0$, then

$$\operatorname{Re}(z/\delta) \geq \frac{m - eC}{1 - e^2}\omega, \qquad |z/\delta| \leq \frac{C}{1 - e}\omega.$$

Proof. Put $w = 1/\delta$. The inequality $|1 - w| \le e|w|$ is equivalent to

$$\left| w - \frac{1}{1 - e^2} \right| \le \frac{e}{1 - e^2}.$$

This closed disk excludes zero and is exactly the image of the original disk under inversion. The minimum of the real-linear function $\mathrm{Re}(zw)$ over a disk of real center c and radius r is $c \,\mathrm{Re}\, z - r|z|$, attained at its support point. The case $e = 0$ is immediate. Substitution proves the identity and both bounds. □

Theorem 3.2 (Accretivity after actual drift division)**.** *Suppose uniformly on the real interval that*

$$\mathrm{Re}(a/H) \ge m_r E, \quad |a/H| \le C_r E, \qquad \mathrm{Re}(Hc) \ge m_\theta, \quad |Hc| \le C_\theta,$$

and $|b| \le \varepsilon\sqrt{E}$*. If*

$$|d_1 + aH_s/H^2| \le e < \min\{1, m_r/C_r, m_\theta/C_\theta\}, \tag{10}$$

then the normalized operator in (8) *has strictly positive radial and angular real margins*

$$\frac{m_r - eC_r}{1 - e^2} E, \qquad \frac{m_\theta - eC_\theta}{1 - e^2},$$

and mixed coefficient at most $\varepsilon\sqrt{E}/(1 - e)$*. The mixed smallness must still fit the unused principal-tensor margin. Bounded ordinary derivatives of* H, H^{-1}, δ *and the original normalized coefficients give bounded derivatives of the new coefficients at every specified finite order.*

Proof. Apply Lemma 3.1 separately with $z = a/H$ and $z = Hc$. The mixed estimate uses $|\delta| \ge 1 - e$. For a derivative, differentiate $\delta^{-1}\delta = 1$ recursively. At any fixed order this is a finite sum of products of derivatives of δ divided by positive powers of δ; the lower bound $1 - e$ bounds every denominator. Apply the product rule to the pulled coefficients. For the principal cross term use Young's inequality $2|\tilde{b}||\xi||\eta| \le \epsilon_r E|\xi|^2 + |\tilde{b}|^2(\epsilon_r E)^{-1}|\eta|^2$ with ϵ_r smaller than the radial margin. This explains the additional mixed-margin requirement rather than asserting accretivity from two diagonal inequalities alone. □

This gives an explicit test for the source's $\delta = 1 + O(E)$ argument. For instance, margins $m_r = m_\theta = 1/2$, bounds $C_r = C_\theta = 2$, and $e = 1/10$ leave each scaled diagonal margin at least $10/33$. The condition $\delta \ne 0$ alone gives no such information: with $a = E$, $c = 1$, $d_1 = 0$, and constant $H = i$, one has $\delta = 1$ but both pulled diffusion coefficients have zero real part. With $H = -1$ they are negative. These are failures of a proposed inference, not admissible source contours. The source's tangent inequalities and normalized derivative bounds are indispensable.

3.1 Fixed row normalization and a nonexact profile

If a nonlinear residual is $F(u)$ and the row factor depends on the unknown, differentiating $m(u)F(u)$ produces

$$D(mF)(u_0)v = m(u_0)DF(u_0)v + (Dm(u_0)v)F(u_0).$$

Thus at a comparison profile with nonzero residual, one cannot identify its linearization with the divided circular linearization unless the row is fixed during variation or the extra term is included. The source fixes the circular row in its height gauge. Our theorem concerns that fixed row and its actual source factor. It makes no assertion about arbitrary parameter-dependent row changes with omitted residual terms.

4 Minimal Compact Observation Enlargement

On $K^\perp$ let $G = (L|_{K^\perp})^{-1}$ in the source Gaussian realization and let

$$\mathcal{V} = C_c^\infty(S) \cap K^\perp, \qquad Q(f,g) = \langle Gf, g\rangle_G.$$

We use the real symmetric pairing first and complexify afterward. The compact orthogonality space is dense in $K^\perp$: subtract a finite compact kernel-dual correction from compact approximants. If $Q(f,g) = 0$ for every $g \in \mathcal{V}$, density implies $Gf = 0$ and then $f = 0$. Thus Q is nondegenerate on the ambient algebraic space $\mathcal{V}$, but it need not be positive. This distinction is central to the source's observation enlargement.

Theorem 4.1 (Minimal balanced completion)**.** *Let $V \subset \mathcal{V}$ be finite dimensional and put $R = \operatorname{rad}(Q|_V) = \{r \in V : Q(r,V) = 0\}$, $\dim R = r$. There is a nondegenerate subspace $W \subset \mathcal{V}$ containing V with*

$$\dim W = \dim V + r. \tag{11}$$

No nondegenerate finite-dimensional subspace containing V has smaller dimension. More precisely, for a complement $V = N \oplus R$, bases $r_1, \dots, r_r$ of R, and suitable $w_1, \dots, w_r \in \mathcal{V}$, the Gram matrix on W is

$$\begin{pmatrix} Q_N & 0 & 0 \\ 0 & 0 & I_r \\ 0 & I_r & 0 \end{pmatrix}, \qquad W = N \oplus R \oplus \operatorname{span}\{w_i\}. \tag{12}$$

The new hyperbolic block has all singular values equal to one in these coordinates. This is not a bound for the physical norms of the chosen compact functions.

Proof. The restriction to any complement N is nondegenerate. Indeed a vector of N orthogonal to N is orthogonal to R as well, hence lies in $R \cap N = 0$. The functionals $Q(r_i, \cdot)$ on $\mathcal{V}$ are independent, by ambient nondegeneracy. Their map to $\mathbb{R}^r$ is therefore onto. Choose $\tilde{w}_j$ with $Q(r_i, \tilde{w}_j) = \delta_{ij}$. Subtract their Q-projections onto N, which exist because Q_N is invertible. This preserves those pairings and makes $Q(N, \tilde{w}_j) = 0$.

Write $C_{ij} = Q(\tilde{w}_i, \tilde{w}_j)$ and define

$$w_i = \tilde{w}_i - \frac{1}{2}\sum_j C_{ji} r_j.$$

Symmetry and $Q(r_i, r_j) = 0$ give $Q(w_i, w_j) = C_{ij} - C_{ij}/2 - C_{ji}/2 = 0$. All cross-pairings with the r_i stay unchanged. They also show that the w_i are independent modulo V. This proves (12) and nondegeneracy. All operations are finite linear combinations of compact smooth functions, so their supports remain compact.

For minimality, suppose W' contains V and is nondegenerate. The map $R \to (W'/V)^*$ induced by $r \mapsto Q(r, \cdot)$ is well defined since $Q(R,V) = 0$. Its kernel is zero by nondegeneracy of W'. Hence $r \le \dim W' - \dim V$, proving (11) is minimal. □

Corollary 4.2 (Prescribed compact sources with no redundant added pairs)**.** *Start with exactly $\dim K$ compact observations whose kernel pairing is invertible. Suppose a prescribed finite list of additional compact sources lies in $\mathcal{V}$ and spans V. Adding observations and multiplier columns for a basis of the space W in Theorem 4.1 makes the full Gaussian augmented map an isomorphism. Among enlargements of V using the pairing Q, precisely r further source/observation pairs suffice and are necessary.*

Proof. In a homogeneous augmented equation, projection of its PDE row onto K first kills the original multiplier coefficients. The remaining source lies in $W \subset K^{\perp}$. Write $u = u_K - Gw$, where w is its multiplier combination. The added observations give $Q(w, W) = 0$, because u_K is orthogonal to W. Nondegeneracy yields $w = 0$. The original observations then kill u_K. For an inhomogeneous right side, the same kernel projection determines the original multipliers; applying G to the remaining $K^{\perp}$ forcing gives a particular solution. The added observations determine w by the invertible Gram matrix, and the original observations determine u_K. These finite operations and the bounded G give a bounded inverse on the stated domain and range. □

The source proves existence of a nondegenerate enlargement. The refinement here gives the exact additional count and removes its self-pairing matrix C by a balanced choice. It reuses ordinary indefinite linear algebra; there is no historical priority claim. Its benefit is fewer auxiliary rows and a transparent finite block. Its cost is that the compact duals may have large physical norms; coordinate balancing does not make those costs disappear.

4.1 An indefinite example

In $\mathbb{R}^3$, let $L = \operatorname{diag}(0, 1, -1)$, with initial kernel observation e_0. On $K^{\perp}$ take

$$q = (e_1 + e_2)/\sqrt{2}, \qquad w = (e_1 - e_2)/\sqrt{2}.$$

Then $Q(q, q) = 0$, $Q(q, w) = 1$, $Q(w, w) = 0$. Adding only the observation q gives a singular augmented system despite linear independence from e_0: the nonzero homogeneous pair $u = -Gq$, $\lambda_q = 1$, $\lambda_0 = 0$ has all observations zero. Adding w produces the unit hyperbolic block and restores invertibility. This finite example isolates the mechanism; it is not a discretization certificate for a shrinker.

5 Full Augmented Schur Residuals

The private core-pat5 proof [2] keeps both the Schur residual and the eliminated interior forcing. We now apply that idea to already normalized PDE operators, not to an unnormalized Jacobi compression. Let $\mathcal{A} : X \to Y$ be a bounded isomorphism between the full weighted Banach spaces, including their old multiplier and observation components. Let $U : \mathbb{C}^r \to Y$ insert new compact PDE columns, and $V : X \to \mathbb{C}^r$ be the new observation map. Put

$$\mathbb{A} = \begin{pmatrix} \mathcal{A} & U \\ V & 0 \end{pmatrix}, \qquad Z = \mathcal{A}^{-1}U, \qquad S = -VZ.$$

Theorem 5.1 (Exact and inexact additional normalization)**.** *The full operator $\mathbb{A} : X \oplus \mathbb{C}^r \to Y \oplus \mathbb{C}^r$ is an isomorphism if and only if S is invertible. For data (f, g) its inverse is*

$$z_0 = \mathcal{A}^{-1}f, \qquad \lambda = S^{-1}(g - Vz_0), \qquad z = z_0 - Z\lambda. \tag{13}$$

Suppose an approximate column action $\widetilde{Z}$ has full residual $E_U = U - \mathcal{A}\widetilde{Z}$ and set $\widetilde{S} = -V\widetilde{Z}$. A sufficient quantitative condition is

$$\sigma_{\min}(\widetilde{S}) > \left\|V\mathcal{A}^{-1}\right\| \|E_U\|. \tag{14}$$

In that case $\|S^{-1}\|$ *is bounded by the reciprocal of the difference in* (14). *For any approximate pair* $(\hat{z}, \hat{\lambda})$, *define its two actual residuals*

$$r = f - \mathcal{A}\hat{z} - U\hat{\lambda}, \qquad s = g - V\hat{z}.$$

Then its errors satisfy exactly

$$e_\lambda = S^{-1}(s - V\mathcal{A}^{-1}r), \qquad e_z = \mathcal{A}^{-1}r - Ze_\lambda. \tag{15}$$

Proof. Elimination of the first row gives $z = \mathcal{A}^{-1}f - Z\lambda$, and the second row becomes $S\lambda = g - V\mathcal{A}^{-1}f$. If S is singular, a nonzero vector λ in its kernel produces a nonzero homogeneous pair $(-Z\lambda, \lambda)$, so invertibility of the full operator fails. This proves both directions and (13).

The exact identity $Z - \widetilde{Z} = \mathcal{A}^{-1}E_U$ gives $S - \widetilde{S} = -V\mathcal{A}^{-1}E_U$. In the declared Euclidean metric on the finite observation coordinates, the least singular value decreases by at most this perturbation norm. Strict positivity gives invertibility of the square S and the stated inverse bound. Finally subtract the full equations for the exact and approximate pairs and repeat the same elimination, yielding (15) without an omitted affine source term. □

For example, with $K = \|\mathcal{A}^{-1}\|$ and $s_0 = \|S^{-1}\|$,

$$|e_\lambda| \le s_0(|s| + \|V\mathcal{A}^{-1}\|\,\|r\|), \qquad \|e_z\| \le K\,\|r\| + \|Z\|\; s_0(|s| + \|V\mathcal{A}^{-1}\|\,\|r\|).$$

These estimates keep the old multipliers inside z. A small matrix $\widetilde{S}$ is useful only after the full-domain action errors E_U have been bounded. Finite samples alone do not provide those bounds.

5.1 One-sided residuals and coordinate changes

Let $B : Y \to X$ be bounded. If $\|I_X - B\mathcal{A}\| < 1$, then $B\mathcal{A}$ is invertible, so $\mathcal{A}$ is injective with closed range and a lower bound. This does not prove it is onto. On $\ell^2(\mathbb{N}_0)$ the right shift T and left shift T^* satisfy $T^*T = I$ but $TT^* = I - P_0$. Thus $B = T^*$, $\mathcal{A} = T$ has zero left residual and fails surjectivity. Conversely the reversed pair has zero right residual and fails injectivity.

If both $\|I_X - B\mathcal{A}\| < 1$ and $\|I_Y - \mathcal{A}B\| < 1$, Neumann series yield

$$\mathcal{A}^{-1} = (B\mathcal{A})^{-1}B = B(\mathcal{A}B)^{-1}.$$

Alternatively a Fredholm index-zero hypothesis plus the left estimate suffices. This is precisely why the source retains its stopped index argument. The finite core-pat5 Neumann proof uses an induced contraction norm and full residual, not merely a spectral radius or a sampled row.

The private stability proof [4] assumes bounded isomorphisms $C : X \to \widehat{X}$ and $D : Y \to \widehat{Y}$. Then

$$\widehat{F}(\hat{u}) = DF(C^{-1}\hat{u}), \qquad \widehat{B} = CBD^{-1}, \qquad \hat{u} - \widehat{B}\widehat{F}(\hat{u}) = C\{u - BF(u)\}.$$

This is exact equivalence. A redundant analysis map with a left inverse is not automatically such an isomorphism onto all coefficient space. Likewise chart multiplication $u = A\tilde{u}$ changes the exit condition to $\tilde{u}_s + (A_s/A)\tilde{u} = g/A$ and creates a zeroth-order term containing A_s/A with no small factor E. Both changes must be charged in the same normed realization.

6 Wider Preliminary-Contour Tolerances

This section concerns the preliminary fixed-radius contours of the source's arrays proposition. All contour constants, scaled overlap lengths, positive terminal radius, and angular orders are fixed before $B \to \infty$. In particular it does not concern the later shrinking exits with $|Br|$ fixed. Choose a cylindrical growth exponent $\alpha > 2.1$ and use the source's forward-compatible weight ρ_B. On the long incoming interval $\rho_B \asymp (1+z)^\alpha$; the bounded scaled bands change it only by constants depending on the fixed contour. Thus

$$1 \le \rho_B \le CB^\alpha. \tag{16}$$

Let X_B and Y_B include the core, conical, compact multiplier and observation norms and the stripwise norms $\sup_I \rho_B(I)^{-1} N_E$. Natural exit data carry $\rho_B(T)^{-1} \|g\|_{T^N_{E(T)}}$.

6.1 The source-specific mixed-gauge mechanism

In the prescribed chart

$$X(t,\theta) = \big(B(\zeta_0 + c_1 u), (R_0 + c_2 u) e_r(\theta)\big),$$

write $a = (\zeta_0 + c_1 u)_t$, $b = (R_0 + c_2 u)_t$, $R = R_0 + c_2 u$. The transversality determinant

$$D = c_2 a - c_1 b = D_0 + (c_2 c_1' - c_1 c_2')u$$

contains no u_t. With $E = B^{-2}$, $d = a^2 + Eb^2$, $L_1 = c_1 a + E c_2 b$, and $T_1 = R^2 d + D^2 u_\theta^2$, the source computes

$$g^{tt} = E/d + L_1^2 u_\theta^2/(dT_1), \quad g^{t\theta} = -L_1 u_\theta / T_1, \quad g^{\theta\theta} = d/T_1,$$

and hence the exact Schur identity

$$g^{tt} - (g^{t\theta})^2 / g^{\theta\theta} = E/d. \tag{17}$$

This algebra explains why radial diffusion is not lost when the full metric is used. It does not by itself give a positive real part after a complex pullback; Section 3 is still needed.

The first traces have sizes $(u, u_\theta) = O(W)$, $u_t = O(BW)$, while the bulk slots have sizes $(u_t, u_{\theta\theta}) = O(W)$, $u_{t\theta} = O(BW)$, $u_{tt} = O(B^2 W)$. The source proves for its full commuted nonlinear remainder

$$\|\mathcal{N}(u)\|_{\text{strip}} \le C(BW^2 + B^2 W^3), \qquad \operatorname{Lip} \mathcal{N} \le C(BW + B^2 W^2), \tag{18}$$

where a difference occupies one slot and a highest second derivative remains in its bulk slot. The trace loss B cannot be dropped. On earlier cylindrical strips its local value is smaller than B; core and conical analytic estimates are bounded and fit the same upper envelope. This is the concrete source input for the next theorem, rather than a generic smoothness assertion.

Theorem 6.1 (Weighted preliminary tolerance)**.** *On the fixed-radius preliminary domains assume the full augmented linear map, with its actual outgoing derivative, has inverse and boundary-lift bounds at most K, independent of B and conical exhaustion. Assume the mixed estimate* (18), (16)*, and the corresponding bounded core/conical estimates hold on the comparison neighborhood. If the total weighted comparison residual, observation discrepancy, and natural exit discrepancy are bounded by εB^{-p}, with fixed $\varepsilon > 0$ and*

$$p > \alpha + 1, \tag{19}$$

then for all sufficiently large B there is a unique correction in a fixed ball of size $C\varepsilon B^{-p}$ in X_B. Its physical strip size is at most $C\varepsilon B^{\alpha - p}$. Constants and the threshold may depend on the fixed data and on ε, not on the conical stop. This uniqueness is only within the declared perturbative ball.

Proof. Include a bounded natural boundary lift in the affine domain, or equivalently use the inverse with the boundary trace as an additional output. For a correction with X_B norm at most r, its physical size on a strip is at most $C\rho_B r$. Divide the first bound in (18) by ρ_B and take the supremum. It gives

$$\|\mathcal{N}(u)\|_{Y_B} \le C(B^{\alpha+1}r^2 + B^{2\alpha+2}r^3). \tag{20}$$

The difference version on the same ball is

$$\|\mathcal{N}(u) - \mathcal{N}(v)\|_{Y_B} \le C(B^{\alpha+1}r + B^{2\alpha+2}r^2)\,\|u - v\|_{X_B}\,. \tag{21}$$

The source's full commuted estimates justify these expressions also for bulk Hessian terms; no product of two uncontrolled second traces was introduced. The bounded core and conical terms are smaller than the displayed envelope for $B \ge 1$.

Let $r = 2K\varepsilon B^{-p}$, enlarging the fixed convention for ε to include the boundary lift. The fixed-point map is the negative inverse applied to the residual plus nonlinear remainder, with the prescribed affine exit data. Its center has norm at most $r/2$. By (21) its Lipschitz constant is at most

$$CK\{2K\varepsilon B^{\alpha+1-p} + 4K^2\varepsilon^2 B^{2\alpha+2-2p}\} \longrightarrow 0.$$

For large B this is at most $1/2$. As $\mathcal{N}(0) = 0$, the same estimate bounds the nonlinear image by $r/2$, so the closed ball is invariant. Completeness and contraction give existence and uniqueness. The weighted envelope gives the physical bound. Finally $BW \le C\varepsilon B^{\alpha+1-p} \to 0$, so the solution remains within the small-trace region in which (18) was derived. This closes, rather than presupposes, that nonlinear hypothesis. Uniformity in the conical stop follows from the assumed uniform inverse and tame constants; the source's exhaustion procedure can then be used. □

Corollary 6.2 (Larger data class for the unchanged comparison)**.** *For the original preliminary comparison, choose*

$$2.1 < \alpha < 3, \qquad \alpha + 1 < p \le 4. \tag{22}$$

Its weighted residual is $O(B^{-4})$, and outgoing and observation discrepancies of size $O(B^{-p})$ give a correction $O(B^{-p})$. In particular $\alpha = 2.2$ permits every $3.2 < p \le 4$.

Proof. The source's cylindrical comparison residual is bounded by $CB^{-4}(1+z)^{2.06}$ and its compact/conical residual by CB^{-4}. Since $\alpha > 2.1 > 2.06$, division by the chosen weight gives $O(B^{-4})$ uniformly. For $p \le 4$ this is $O(B^{-p})$, so Theorem 6.1 applies. The interval in (22) is nonempty exactly when $\alpha < 3$. □

If one prescribes B^{-p} data with $p > 4$ but keeps the same comparison, the total discrepancy is only $O(B^{-\min\{p,4\}})$; the theorem cannot promote it to $O(B^{-p})$. The abstract theorem allows $p > 4$ only when that stronger residual hypothesis is separately proved. At the equality $p = \alpha + 1$, small constants might give a contraction, but scale decay alone gives none; no endpoint claim is made here.

At a fixed-radius outer exit, $E(T) \asymp B^{-2}$ and $\rho_B(T) \asymp B^{\alpha}$. A sufficient smooth outgoing derivative tolerance is therefore

$$\|g\|_{H^{h+1}} \le cB^{\alpha+1-p}, \tag{23}$$

because its weighted natural trace norm is at most $CB^{-\alpha-1}\,\|g\|$. For $\alpha = 2.2$, $p = 3.5$ yields physical correction $O(B^{-1.3})$ and permits $O(B^{-0.3})$ smooth exit discrepancy, whereas $p = 4$ yields $O(B^{-1.8})$ and $O(B^{-0.8})$, respectively. These are sufficient tolerances, not lower bounds or claims about every rough trace. Holomorphic data on an outer finite-parameter ball retain holomorphic solutions by uniform contraction; parameter derivative bounds on an inner ball pay the usual inverse-distance powers.

6.2 What this enlargement does not preserve

The source's later mean evaluation needs error $o(B^{-4})$ after dividing by an inner z^2 and matching the actual stationary solution to finite jets. Its action calculation needs additive $o(1)$ accuracy in Z_*^2, not merely a relative pole estimate. Larger preliminary data can alter those quantities. The source's prescribed data and their $p = 4$ uniqueness balls remain the inputs to those later statements. Likewise the shrinking-height nonlinear coefficient is $E^{-1/2}|r|^{-2} \asymp B/|r|$, not just B. Its separate mean and oscillatory weights are not covered by (16). We do not transfer our extension to that regime.

7 Shielding and the Unchanged Global Obligations

7.1 A shifted exit is a Robin exit

The source's shielding proof acts on a fixed overlap with $E \asymp B^{-2}$. For $d = c_1 B^2$ with Ed small and $v = e^{d(t-T)}w$, the model becomes

$$e^{-d(t-T)}(\partial_t - E\partial_t^2 - c\partial_\theta^2)v = -Ew_{tt} + (1 - 2Ed)w_t - cw_{\theta\theta} + (d - Ed^2)w. \tag{24}$$

The true exit $v_t(T) = g$ is $w_t(T) + dw(T) = g$. For real frozen coefficients its terminal energy coefficient is

$$Ed + \tfrac{1}{2}(1 - 2Ed) = \tfrac{1}{2}.$$

If λ_f is the original fast exponent, the transformed exponent is $\lambda_f - d$ but the Robin lift divides by $(\lambda_f - d) + d = \lambda_f$, not by $\lambda_f - d$. Replacing the Robin exit by a Neumann exit would change the problem and can create an artificial small denominator.

The source keeps $D = \partial_t + d$ intact in the shifted norm

$$d\,\|w\| + \|Dw\| + \|w_{\theta\theta}\| + \left\|ED^2 w\right\| + \left\|\sqrt{E}Dw_\theta\right\|.$$

Relative second-order perturbations are measured by $\delta a/E$, $\delta b/\sqrt{E}$, and δc; lower coefficients containing a background second derivative remain in the appropriate L^2 module. A graph of size W produces an operator perturbation controlled by $BW + (BW)^2$, provided the background and reference chart are the stated ones. This is stronger information than absolute coefficient closeness. It leads to an inner overlap error $B^M e^{-cB^2}$. Only then is the polynomial physical inverse used to repair a splice. The estimates here inherit this shielding theorem, not a new proof of its full variable-coefficient PDE estimate.

7.2 Formal jets, actual arrays, and action phases

The source constructs S_b and polynomial opening jets in x in real spatial charts; the formal action belongs to $\mathbb{R}\{b\}[[x]]$. At each degree the unknown remains on the left of the normalized Jacobi equation and lower-degree sources occupy product weights with one highest bulk derivative. Finite-order polynomial growth does not imply convergence of the opening series. The opening coefficient is $\beta = M(b)x$ with an invertible first-jet map; this is not a global analytic stationary branch in x.

For a nonzero formal critical arc, its opening scale is $B = |s|^{-k/2}$. The contour construction supplies strict diffusion margins in one real parameter. Actual finite arrays are holomorphic in finite

parameters with prescribed outer/inner ball reserve. Their action variation retains both boundary momentum and moving-boundary terms even when outgoing derivatives are prescribed. Compact multiplier terms vanish only for fixed observations. Nonadverse boundary heights may still vary and their far momenta must be estimated.

The source obtains Gevrey bounds by telescoping consecutive real stops with exponentially small differences on disks of radius comparable to $R^{-2/k}$. Identification of each formal coefficient uses one sufficiently high finite Taylor order and real comparison before analytic continuation. Division by s^q first subtracts the lower Taylor polynomial; a merely small stopped function need not be divisible. The constrained formal substitution retains all constraints using cofactor derivations with controlled determinant valuations. This is not unrestricted differentiation of an ideal. Regular implicit contraction and a sectorial flatness argument then compare the two actual actions at common parameters. These are independent obligations beyond any local inverse theorem.

On a collapsing end write $w = R^2/2 - 1$, $t = \log z$. Angular curvature has the exact divergence form

$$\frac{4(1+w)w_{\theta\theta} - 4w_\theta^2}{4(1+w)^2 + w_\theta^2} = 2\partial_\theta \arctan\left(\frac{\partial_\theta \log(1+w)}{2}\right).$$

Its mean vanishes, but that does not justify dropping the axial/angular products in the full equation. Keeping them gives the source's first mean jets

$$\begin{aligned}
\langle w_{[0]} \rangle &= O(z^{-4+\eta}),\\
\langle w_{[1]} \rangle &= \beta(z^2 - 2) + O(|x| z^{-2+\eta}),\\
\langle w_{[2]} \rangle &= (-4\beta^2 \log z + c_2(b,x))z^2 + O(|x|^2 z^{1+\eta}).
\end{aligned}$$

For $L_0 = \partial_z^2 - (z/2)\partial_z + 1$ and $h_0 = z^2 - 2$,

$$(h_0^2 e^{-z^2/4} v')' = h_0 e^{-z^2/4} L_0(h_0 v).$$

Polynomial growth removes the fast flux at infinity. The particular solution $-4\beta^2 z^2 \log z$ has image $2\beta^2 z^2 - 8\beta^2 \log z - 12\beta^2$. This identifies the resonant logarithm, while the remaining flux integral defines the real quadratic coefficient c_2. These formulas concern finite jets and controlled remainders, not a summed formal profile.

Actual inner mean matching uses a high Taylor order at $z_1 = B^{1-\epsilon}$, with a farther real stop $z_2 = B^{1-\epsilon/2}$. The forced residual is first solved in its own growing weight; a separately shielded boundary lift handles the remote trace. Keeping the compact multiplier and observation equations excludes a compensating slow homogeneous mode. Along the mean profile a second derivative is integrated by parts with its genuine first trace. This gives the needed $o(B^{-4})$ primitive error and the phase expansion

$$Z_*^2 = -\frac{1}{\beta} - 8\log Z_0 + \frac{c_2(b,x)}{\beta^2} + 2 + 2\log 2 + o(1), \quad Z_0 = (-\beta)^{-1/2}. \tag{25}$$

A relative $1 + o(1)$ approximation to Z_*^2 would be insufficient for its Gaussian exponential.

7.3 Why a nonzero local jump is a separate theorem

After matching, the exact radial action has prefactor

$$\frac{8\pi}{Z_*^2} e^{-Z_*^2/4} y e^{h-(h^2+y^2)/Z_*^2} \sqrt{1+l^2}\, dy, \qquad y = Z_* r/2, \quad h = (Z_* - z)Z_*/2.$$

At fixed large exterior cutoff Y_0, one first takes $B \to \infty$. The refined mean/oscillatory bounds eliminate oscillation and give the radial translator equation

$$h' = l, \qquad l' = (1 + l^2)(1 - l/y).$$

This is an exterior limit of complex stationary boundary problems, not an assumed translator cap in the evolving real flow. Then $Y_0 \to \infty$ controls the relative action error.

For $q = 1 - l/y$ and $\tau = y^2/2$, direct substitution gives

$$q_\tau + q = 2q^2 - q^3 + \frac{1 - 2q}{2\tau}.$$

The source constructs lateral solutions by Volterra contraction toward upper-left and lower-left infinity. In $x = y^{-2}$ their common formal coefficients satisfy

$$b_1 = 1, \quad b_{n+1} = \sum_{j=1}^{n} (2n - j + 1) \sum_{i_1 + \cdots + i_j = n} \prod_{\ell=1}^{j} b_{i_\ell}, \quad 2^{n-1}(n-1)! \le b_n \le 3^{n-1}(n-1)!.$$

Positive coefficient growth alone is not the nonzero-jump proof. The source also proves factorial remainders, constructs an inverse Laplace continuation under the supposition of equal lateral solutions, and contradicts the positive-axis singularity of the Borel series. Only then is the difference equation used to deduce $l_+ - l_- = c_0 y^3 e^{-y^2/2}(1 + O(y^{-2}))$, $c_0 \neq 0$.

For the limiting momentum $\mathcal{P} = y e^h l/\sqrt{1 + l^2}$ one has

$$\mathcal{P}' = y e^h \sqrt{1 + l^2}, \qquad d\mathcal{P} - \mathcal{P}\, dh = y e^h (1 + l^2)^{-3/2} dl \quad (y \text{ fixed}).$$

At finite Z_* the corrected momentum instead differentiates to $(1 - 2h/Z_*^2)$ times the corrected Lagrangian; it is not an exact primitive. The source therefore takes exact finite-scale first variations before the translator limit. This boundary cancellation, normalized compatibility of longer and shorter solutions, and common outward orientations yield a common nonzero constant multiplying each $Z_*^{-2} e^{-Z_*^2/4}$.

Finally (25) is used on a generic nearby ray. Distinct negative-power Laurent principal parts separate exponentially. Equal principal parts have real constant phase differences and hence positive limiting ratios $e^{-d_j/4}$. This excludes cancellation in the dominant class. It contradicts the independently derived action upper bound for a formal critical opening arc. Arbitrary complex end phases would not satisfy this argument. None of the four local theorems here replaces any of these steps.

8 Finite-Jet to Flow: A Conditional Reconstruction

8.1 Finite power and localized gradient

The source's no-critical-opening-arc theorem feeds a real formal Nullstellensatz. For each opening coordinate it obtains

$$x_i^{2m_i} + \sum_j h_{ij}^2 = \sum_j a_{ij} \partial_j \mathcal{F}.$$

The exponents are fixed first. Reduction modulo $(x)^{N-1}$ leaves finitely many coefficient equations in b; the source invokes analytic approximation for that finite system, not convergence of the full formal action. For one high order N, positivity over the reals gives

$$|x|^m \le C_N\big(|\nabla f_N| + |x|^{N-1}\big), \qquad m = 2\max_i m_i. \tag{26}$$

This is inherited together with the algebraic theorems used for it. It is not a consequence of invertibility of a finite observation matrix.

Let M be an entire single small-height, small-slope graph over S_b out to radius R, with the source's uniform area-ratio bound and compact observations. Additional sheets are excluded by this hypothesis. Set

$$\mathcal{D} = \|\Phi_M\|_{L^2_G(M)} + e^{-(1-\eta)R^2/8}.$$

The actual graph need not have a small Hessian: coefficient subtraction keeps its principal coefficients on the unknown difference and uses a small Hessian only on the finite Taylor comparison. The source obtains

$$\|\chi_R(u - v_N)\|_{H^2_{1+\delta}} + |\lambda_N| \le C(\mathcal{D} + |x|^{N+1}), \qquad |\nabla f_N| \le C(\mathcal{D} + |x|^N).$$

Combining with (26) and absorbing the higher power gives $|x| \le C\mathcal{D}^{1/m}$. Choose $N - 1 > 4\max\{m, 1\}$. The finite-dimensional analytic gradient inequality for this single f_N then applies. The action comparison still needs a weight interpolation, since the reference surface has infinite unweighted area.

Explicitly choose $0 < \theta < \min\{\theta_0, 1/2\}$, $\alpha_0 < 1$ with $2\alpha_0(1-\theta) > 1$, then $\delta > 0$ with $a_* = \alpha_0(1+2\delta) + (1-\alpha_0)/2 < 1$. For polynomially weighted first graph differences U, Hölder gives

$$\int U^2 e^{-a_*|Y|^2/4} \le \left(\int U^2 e^{-(1+2\delta)|Y|^2/4}\right)^{\alpha_0} \left(\int U^2 e^{-|Y|^2/8}\right)^{1-\alpha_0} \le C\mathcal{D}^{2\alpha_0}.$$

Second variation and Gaussian tails then give $|F(M) - f_N| \le C\mathcal{D}^{2\alpha_0}$. Therefore

$$|F(M) - F(S)|^{1-\theta} \le C_\eta \mathcal{D}. \tag{27}$$

All exponents are chosen before the arbitrary localization loss η; the constants, graph tolerance, and minimum radius may depend on it. If there are no opening variables, the finite-power step is omitted, but actual graph comparison and Gaussian tails remain necessary.

8.2 Delayed radii and a finite-prefix proof

Here is a complete reconstruction of the scalar summation that uses (27), conditional on the source's geometric propagation. In rescaled time τ, let $\mathcal{E} \ge 0$ satisfy $-\mathcal{E}' = d^2$. Set

$$a_j = (\mathcal{E}(j) - \mathcal{E}(j+1))^{1/2}, \quad r_j = \sqrt{8\log(1/a_j)}, \quad R_j = \min\{Q^j R_{\text{in}}, \min_{i \le j} Q^{j-i} r_i\},$$

where $Q > 1$, $a_j < 1$, and $r_j = \infty$ at zero. Then $R_{j+1} = \min\{QR_j, r_{j+1}\}$. Assume the propagated localized estimate supplies

$$\mathcal{E}(j+1)^{1-\theta} \le C(a_j + e^{-(1-\eta)R_j^2/8}). \tag{28}$$

Choose $0 < \eta < c < \theta < 1/2$, put $p_0 = (\theta - c)/(1-c)$, $\kappa = (1-\eta)/(1-c) > 1$, $q = Q^2$, $h = \kappa/8$, and $\delta_0 = \sqrt{\mathcal{E}(0)} < 1$. Define

$$\rho(\delta_0) = \sum_{\ell \ge 1} \delta_0^{\kappa q^\ell - 1}, \qquad B_0(R) = \frac{e^{-hR^2}}{1 - e^{-h(q-1)R^2}}.$$

Proposition 8.1 (Conditional prefix bound)**.** *If $\rho(\delta_0) < 1$, every available finite prefix obeys*

$$\sum_{j=0}^{n} a_j \le \frac{C\mathcal{E}(0)^{p_0} + B_0(R_{\mathrm{in}})}{1-\rho(\delta_0)}. \tag{29}$$

Proof. Call j good when the Gaussian term in (28) is at most a_j^{1-c}. Raising the resulting inequality to power $1/(1-c)$ gives $\mathcal{E}(j+1)^{1-p_0} \le C_1 a_j$. Write $u = \mathcal{E}(j)$, $v = \mathcal{E}(j+1)$. If $v \ge u/2 > 0$, concavity gives $u^{p_0} - v^{p_0} \ge p_0 u^{p_0-1} a_j^2 \ge c_1 a_j$, using $u^{1-p_0} \le 2^{1-p_0} C_1 a_j$. If $v < u/2$, then $u^{p_0} - v^{p_0} \ge (1-2^{-p_0})u^{p_0} \ge c_2\sqrt{u} \ge c_2 a_j$, since $p_0 < 1/2$ and $u \le 1$. Thus good terms sum to at most $C\mathcal{E}(0)^{p_0}$; zero terms contribute nothing.

For a bad index,

$$a_j < e^{-hR_j^2} = \max\{e^{-hq^j R_{\mathrm{in}}^2}, \max_{i<j} a_i^{\kappa q^{j-i}}\}.$$

Its own positive term $a_j^{\kappa} < a_j$ cannot realize this upper bound. Consequently $a_j \le e^{-hq^j R_{\mathrm{in}}^2} + \sum_{i<j} a_i^{\kappa q^{j-i}}$. Sum over the finite prefix. Each power is at most $a_i \delta_0^{\kappa q^{j-i}-1}$, so the double sum is at most $\rho(\delta_0)\sum_{i=0}^{n} a_i$. Also $q^j \ge 1 + j(q-1)$ bounds the initial-radius sum by $B_0(R_{\mathrm{in}})$. Absorption gives (29). Finally $\rho(\delta_0) \le \delta_0^{\kappa q-1}/(1-\delta_0^{\kappa(q-1)}) \to 0$, so the hypothesis is available for sufficiently small initial energy. □

The displayed denominator records the unused contraction margin in the source's argument; taking $\rho \le 1/2$ recovers its factor two. This reconstruction does not prove its geometric premise. The source obtains complete-graph propagation from graphical pseudolocality, positive-time interior smoothing, tubular separation, and avoidance, not a pointwise graph at one patch. Selected times lie in $[j+0.2, j+0.4]$, so adjacent delays lie in $[0.8, 1.2]$. Chart changes cost $C(1+R)|b_{j+1} - b_j|$ and need both r_j and r_{j+1}. A final incomplete bootstrap interval is propagated over the fixed two-unit rescaled horizon without looking at a future energy drop. Positive-delay smoothing is used at every selected endpoint. Once these obligations hold, (29) implies finite total motion; observation convergence fixes the ambient frame, while the delayed radii tend to infinity.

The exact identity

$$M_s^{\lambda} = \sqrt{-s}\, M(2\log\lambda - \log(-s))$$

converts all-time rescaled convergence to (1). Spacetime weak convergence and a countable determining family give equality of tangent weights almost everywhere in time. Under the source's all-endpoint convention, subtracting the smooth comparison's equality from the Brakke inequality makes the difference of each nonnegative test integral nonincreasing and zero almost everywhere. Bracketing any interior time by good times makes it zero there too. The endpoint convention is what removes exceptional-time representatives; it is not supplied by an operator inverse.

9 Advantages, Costs, and Further Work

The direct improvement is a larger preliminary data class, with an explicit scale condition and a proof in the source's full weighted norms. The drift result gives a sharp scalar disk margin rather than a qualitative nonzero-divisor test. Minimal observation completion saves exactly the otherwise unnecessary auxiliary pairs and exposes indefiniteness. The Schur result permits reuse of a valid normalized inverse while retaining every source, observation, and multiplier residual. These benefits are local and source-specific; none is a universal speedup or a new general theory of tangent flows.

The costs are equally concrete. Compact dual functions can be badly conditioned in physical norms. Full-domain column residuals are harder to bound than finite matrix residuals. Increasing the preliminary tolerance weakens the actual physical accuracy and can destroy downstream phase matching. The weighted estimate still relies on uniform strip estimates, graph compatibility, the Gaussian augmented inverse, and a stopped index-zero argument. Constants depend on the shrinker, finite charts, contour margins, and fixed derivative orders; no numerical threshold for an arbitrary surface is supplied.

Useful future extensions would require additional proofs. A computable contour validation would need interval bounds for the full normalized coefficient modules and trace maps, not a mesh of sample signs. An optimized observation design would minimize physical dual norms subject to the indefinite Gram constraint; the unit block alone solves only the algebraic conditioning problem. Extending the tolerance class into shrinking heights would need a revised mean/oscillatory weighted contraction and a fresh phase-error budget. Improving the global theorem would require preserving the finite-jet real-radical argument, localized gradient exponent, exact boundary momentum, and noncancellation after all such changes. A formalization would have to expose these analytic interfaces and their hypotheses; checking finite algebra would be only one component.

9.1 Private proof reuse and contribution level

The actual core-pat5 Neumann, Schur-residual, and low-rank-update proofs [2] supply the discipline of typed inverse actions, noncommutative ordering, and retained interior sources. TR8 [3] supplies explicit operator realizations, boundary lifts, and the distinction between an exact inverse reformulation and a contraction theorem. The stability section [4] supplies the bounded-isomorphism transfer and residual-ball hypotheses. They are used here through explicit equations, not merely by renaming ordinary mathematics as Ram calculus. The additional value is their synthesis at M683's mixed-gauge, compact-augmentation, and finite-contour interfaces.

The assessment is C3, independently of any catalogue applicability level. It reflects the proved conditional tolerance enlargement and quantitative local refinements, not an independent proof of the source's global theorem. No priority claim is made for Schur complementation, indefinite completion, disk inversion, Banach contraction, or the classical uniqueness literature.

10 Verification and Read-Scope Boundaries

The analytic proofs above are the mathematical justification. The accompanying bounded test script checks exact rational or symbolic identities for drift pullback, the inversion disk, mixed-gauge Schur cancellation, hyperbolic balancing, source-dependent Schur elimination, shifted Robin terms, and tolerance exponents. Finite floating-point examples exercise complex disk margins and inexact augmented residual transport. Such tests can reveal transcription errors; they prove no interval-uniform estimate, PDE surjectivity, infinite-dimensional frame bound, action-jump nonvanishing, or global uniqueness claim.

The source reading is complete for the fifteen mathematical TeX/bibliography files listed in Section 1. Private reading is deliberately narrower: the stated proof ranges in core-pat5, TR8, and the TR1 stability section, not the entire confidential corpus. Public literature checking covers the cited primary records and their stated scopes, not independent reproofs of their full articles. Native compilation, physical PDF export, every-page visual inspection, exact test results, source hashes,

and final deliverable hashes are recorded in the separate local audit receipt. They do not change the inherited status of the global theorem. All private material remains local.

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